APPLICATION OF A MECHANICAL HEART AND LUNG APPARATUS TO CARDIAC SURGERY

JOHN H. GIBBON, Jr., M.D.
Philadelphia, Pennsylvania

IT IS A PLEASURE to be here and to talk about a subject in which I have been interested for many years The ultimate objective of my work in this field has been to be able to operate inside the heart under direct vision From the beginning, I have not only been interested in the substitution of a mechanical device for the heart, but also for the lung We have always considered congenital abnormalities of the heart the most suitable lesions for operative repair Many of these abnormalities are septal defects In the presence of a septal defect, shunting the flow of blood around one side of the heart with a pump, will not provide a bloodless field for operative closure of the defect An apparatus which embodies a mechanical lung, as well as pumps, enables you to shunt blood around both the heart and lungs, thus allowing operations to be performed under direct vision in a bloodless field within the opened heart Furthermore, an apparatus which embodies a mechanical lung enables you to provide partial support to either a failing heart or failing lung where a major operative procedure is not contemplated Such an apparatus can also be used as an adjunct during the course of a major operative procedure This partial support of the cardiorespiratory functions consists in removing venous blood from some peripheral vein continuously, oxygenating the blood and getting rid of the carbon dioxide in it and then injecting the blood continuously in a central direction in a peripheral artery. Of course, such partial circulation, or cardiorespiratory support, requires the use of a mechanical lung in the circuit

I shall not describe in detail the entire apparatus I shall merely discuss six aspects of the problem which I consider of fundamental importance Four of these concern the apparatus itself, and two concern problems which arise on opening the heart and operating within it under direct vision

The first feature of a mechanical heart-lung apparatus is a suitable pumping mechanism to move the venous blood from the subject, through the apparatus, and back into an artery of the subject There is no real problem about a pumping apparatus There are many ways of moving blood through tubing without producing significant amounts of hemolysis We have used for many years a roller type of pump which does not contain any internal valves Such pumps are extremely simple Because of the absence of valves, the blood circuit is easy to clean and there are no stagnate regions where fibrin might be apt to form There are many other advantages in this type of pump such as the simple and rapid control of the rate of blood flow The pumps cause no significant hemolysis In human patients in which we have used the apparatus, hemolysis has always been well below 100 mg of free hemoglobin per 100 ml of plasma In animal experiments, hemolysis is similarly minimal

The second main feature of a mechanical heart-lung apparatus is the mechanical lung itself This presents far more difficulties than pumping blood I am sure that the most efficient apparatus for performing the functions of the lung has not yet been devised Our present mechanical lung, however, provides a reasonably satisfactory working solution to this problem The mechanical lung performs the gas exchange required for respiratory function by filming blood on both sides of screens which have a somewhat larger mesh than ordinary fly screens These screens are made of stainless steel wire and are suspended vertically and parallel in a plastic chamber As the blood flows over these screens, it takes up oxygen and gives off carbon dioxide It should be remembered that it is equally important to remove carbon dioxide from the blood as it is to add oxygen It is easy to observe that sufficient oxygen is being picked up in the apparatus. as the blue blood entering the oxygenator becomes red as it leaves. This can be determined more accurately, of course, by intermittent sampling or by continuous reading with a Wood cuvette and oximeter On the other hand, there is no way of estimating the carbon dioxide tension by observing the color of

Presented in the Symposium on Recent Advances in Cardiovascular Physiology and Surgery, University of Minnesota, Minneapolis, September 16, 1953

Dr Gibbon is Professor of Surgery and Director of Surgical Research, The Jefferson Medical College, Philadelphia, Pennsylvania

the blood We have solved this problem by reading continuously the pH of the blood as it leaves the oxygenator As there is no significant increase in fixed acids in the blood in the course of these experiments, the pH changes are due practically entirely to changes in carbon dioxide tension We have an automatic control which keeps the carbon dioxide tension at the desired normal level and which is operated by any change in the continuously recorded pH level of the blood.

A third important feature of any apparatus which temporarily performs the function of the heart and lungs is constancy of fluid volume. The apparatus should at all times hold a constant volume of blood at any rate of blood flow. If the apparatus is not designed so as to hold a rigid volume of blood at all rates of blood flow, blood might accumulate in the apparatus with consequent depletion of the subject's vascular system and a dangerous drop in the subject's blood pressure. Similarly, if the apparatus should hold less blood at any time, there would be an excessive amount of blood in the subject's vascular system. Obviously the tubing in the blood circuit will always hold a constant amount of blood. There are two places in the circuit, however, where the blood volume might vary One is in the blood reservoirs at the bottom of the plastic chambers which draw venous blood from the subject, and the other is in the thickness of the film of blood on the screens in the oxygenator. Rigid control of the volume of blood at the bottom of these plastic chambers has been obtained by an electronic device invented by Dr B. J. Miller in our laboratory. This electronic device senses the level of the blood in these chambers and automatically operates the pumps which draw blood from the chambers. Thus when the level of blood tends to rise, the pump automatically operates at a faster rate. When the level falls the pump automatically is slowed. This electronic circuit has proven eminently satisfactory and maintains a rigid volume of blood in these chambers.

In the second place where blood might accumulate, on the screens of the oxygenator, a very simple way to avoid such an increase is by inserting an additional pump in the circuit which draws not only from the bottom of the mechanical lung but also from the tubing carrying venous blood from the subject. This additional pump operates at a rate which is always greater than the rate of blood flow from the venae cavae of the subject As this pump operates at a fixed rate, the thickness of the blood on the screens in the oxygenator does not vary.

The fourth important requirement of any mechanical heart and lung apparatus is that the apparatus should remove all of the blood returning to the heart through the venae cavae and yet should not apply too great a negative pressure to the orifices of the cannulae because the venae cavae would then be collapsed around the orifice of these cannulae We have found that the simplest way of obtaining such a smooth blood flow from the cavae is to interpose a suction chamber between the pump and the cannulae in the venae cavae The degree of negative pressure can be easily regulated in this chamber and a smooth uniform flow of blood is easily obtained.

In summary, then, every mechanical heart-lung apparatus devised to take over temporarily the entire functions of the heart and lungs must comprise four essential features First, there must be a good method of pumping blood through the circuit which does not cause hemolysis, which can be quickly and easily adjusted to varying flow rates and finally which enables the blood circuit to be easily and thoroughly mechanically cleaned Second, the mechanical lung must not only fully saturate the blood with oxygen but must maintain the carbon dioxide tension of the blood at a normal level The latter requirement may be taken care of by an automatic apparatus which continuously reads the pH of the blood leaving the mechanical lung and adjusts the carbon dioxide tension accordingly Third, the apparatus must hold a constant amount of blood at all rates of blood flow. This can be accomplished by electronic control of the pumps removing blood from plastic chambers so that the pumps operate always to maintain a constant level of blood in the reservoirs at the bottom of the chambers The thickness of the blood film on the screens in the oxygenator is kept constant by an extra pump which always circulates a constant flow of blood over the screens Fourth such an apparatus must be able to remove smoothly all the venous blood returning to the heart through the venae cavae without collapsing these veins We have found the simplest way of accomplishing this is to interpose a negative pressure chamber between the pump and the cannulae in the venae cavae.

There are two problems concerning operations upon the open heart which merit discussion. The first consists in the disposal of the blood returning to the chambers of the heart even though all the blood flow from the venae cavae is diverted to the apparatus The second problem is the avoidance of air embolism when the heart is opened.

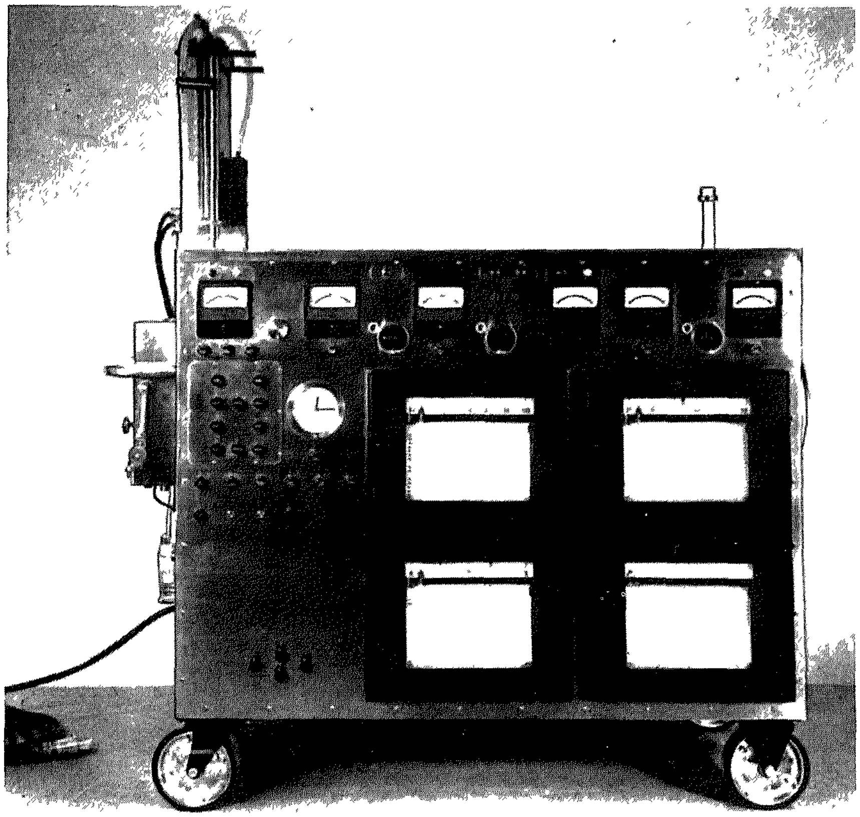

Fig 1 Front view of apparatus showing the recording and control instruments and the lung suspended above the cabinet on the left

The first problem is not difficult to solve When the entire functions of the heart and lungs are taken over temporarily by the mechanical heart-lung apparatus, the myocardium continues to receive its normal flow of oxygenated blood by way of the coronary arteries This blood is returned to the interior of the heart by way of the coronary sinus and the Thebesian veins This blood must be disposed of so that the operative field can be clearly visualized when the heart is opened We have accomplished this by aspirating this blood into a special plastic chamber in which any air aspirated is dissipated The blood collects at the bottom of this chamber and is pumped back into the main extracorporeal circuit free of air

Air embolism must be avoided when the heart is opened. If there is no septal defect, operations upon the right side of the heart can be performed without any great danger from air embolism It is easy to flood the chamber of the heart with blood or salt solution after the operation is completed so as to avoid air embolism Small amounts of air in the pulmonary arteries are probably not significant On the other hand, operations on the left side of the heart, or on the right side of the heart in the presence of a septal defect, present a real problem in the prevention of air embolism into the ascending aorta The immediate result of such air embolism is usually blockage of the coronary arteries with ventricular fibrillation and death Our solution of this problem is to insert a small plastic catheter through a stab wound in the apex of the left ventricle. Suction is applied to this plastic catheter during the course of the open cardiotomy so that any air or blood entering the left ventricle takes the path of least resistance out through this plastic catheter instead of being

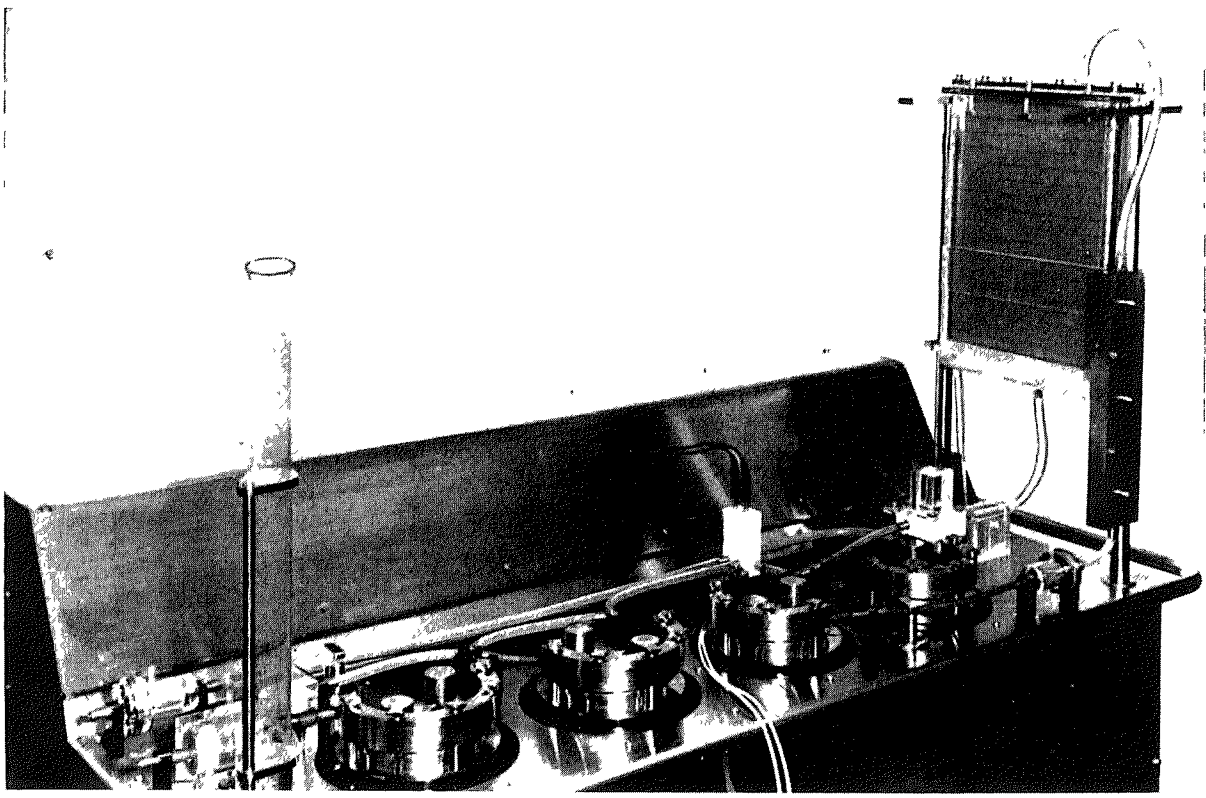

Fig 2 Oblique rear view of apparatus showing the rotary blood pumps and the battery-type screen lung suspended above the cabinet on the right side.

ejected into the aorta This plastic catheter is also connected to the debubbling chamber which receives the cardiac venous blood aspirated from inside the heart Thus this blood is also returned to the circuit, and so to the subject, after the air bubbles are removed After the wound in the heart is closed, and no further bubbles of air appear in the tubing connected to this catheter, it is removed and the small stab wound in the left ventricle closed by suture Since employing this method in both animals and patients, we have had no instances of air embolism.

Figure 1 shows the front view of the apparatus which I have described, with the recording and control devices on the front panel Figure 2 shows the rear view with the roller type pumps on top of the cabinet Two rollers on a revolving arm pass over the rubber tubing which is clamped in a semicircular position The rollers move the blood through the tubing and the blood cannot flow back because there is always a roller compressing the tube To the right of Figure 2 is the mechanical lung which consists of vertical screens suspended in parallel in the plastic case. Blood passes onto the screens through slits at the top of the lung The blood collects in the bottom of the plastic case as it leaves the screens The pump returning the oxygenated blood to the subject is automatically controlled by the electronic device which senses the level of the blood at the bottom of the plastic case, through which the blood passes The lucite block near the oxygenator contains glass and calomel electrodes which continuously measure the pH of the blood. The filter consists of a screen with wires 150 microns in diameter and a 300-micron mesh We do not know whether such a filter is necessary before returning the blood to the subject However, we regard it as a good safety precaution in human patients The tube at the end of the apparatus returns the blood to the patient through a cannulae directed in a central direction in an artery In human patients we employ the central end of the divided left subclavian artery The other two tubes are connected with cannula which are inserted into the superior and inferior venae cavae

Oxygen is blown over the screens suspended in the plastic case Thus the blood film on the screens is exposed to an atmosphere of pure oxygen The lung shown in Figures 1 and 2 has six screens. We have a larger mechanical lung with eight screens which are of longer length and which we have used on adult patients An additional pump draws blood both from the bottom of the oxygenator and from the tubing containing the venous blood coming from the patient The pump maintains a constant rate of flow through the oxygenator so that there is no variation in the thickness of the films on the screens. The electronic control circuit maintains the blood level in the bottom of the plastic case constant at all rates of flow The pump removing the blood from this chamber is controlled by this electronic device.

There are two plastic negative pressure chambers One of them collects the blood from the venae cavae and the other (Fig 3) is the debubbling chamber in which the blood from the cardiac veins and from the left ventricle is collected As the blood passes down the sides of this plastic chamber any bubbles of air are dissipated Last spring we reported the successful repair of interatrial septal defects in animals using a flap of pericardium We have been prepared to use such a flap of pericardium in human patients We found, however, as Swan has, that it is quite easy to close such defects with a continuous suture in the open heart under direct vision and that consequently a pericardial graft is not needed

The inferior vena cava is cannulated by a "tygon" tube passed into the inferior vena cava by way of the right atrial appendage The superior vena cava is cannulated by a "tygon" tube passed through a stab wound in the right atrial wall Oxygenated blood from the apparatus is pumped into the aorta through the divided central end of the left subclavian artery Ligatures passed around the superior and inferior venae cava are tied over the enclosed cannulae This diverts all the venous blood to the extracorporeal blood circuit. In addition to closing atrial septal defects in dogs with the pericardial graft, we have successfully closed interventricular defects in dogs by direct suture. The defect is exposed by an incision in the anterior wall of the right ventricle parallel to the left anterior descending coronary artery

There has been a progressive decline in operative mortality in successive series of animals operated upon in our laboratory. Three years ago we had an 80 per cent mortality In the most recent series this mortality has declined to 12 per

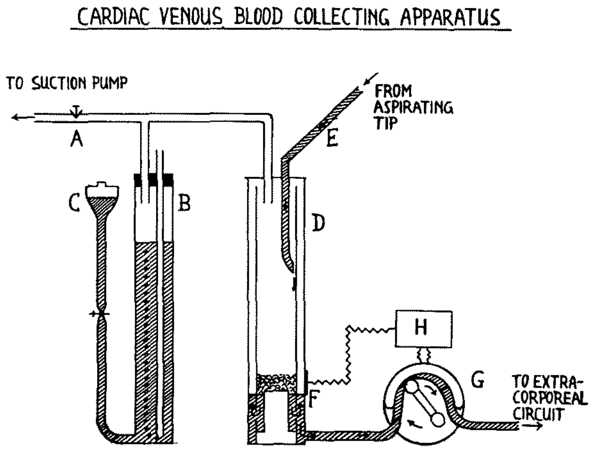

Fig 3 Diagram of the cardiac venous blood collecting apparatus The diagram is self-explanatory The rotary pump G returns the blood to the main extracorporeal circuit after it has dissipated its bubbles in the inner chamber of cylinder D

Reprinted from The Medical Clinics of North America (Volume 37, page 1615) with permission of the publishers

cent By operative mortality we mean any death which could be attributed to the operation and any death occurring the first month after operation. We have successfully temporarily taken over the entire blood flow through the heart and lungs of a dog for as long as one hour and forty minutes with prolonged survival in a healthy condition.

The average saturation of the venous blood entering the apparatus is 63 per cent. We regard a normal saturation of venous blood with oxygen as the best indication of an adequate blood flow to the tissues If the saturation of venous blood with oxygen falls to low levels it is obvious that there is an inadequate blood flow to the tissues We have been successful in maintaining the pH in a normal range. The hemolysis in this group of experiments averaged 35 mg of free hemoglobin per hundred ml of plasma You are all aware that this is an insignificant amount of hemolysis and that a similar degree of hemolysis can occur if blood is forced rapidly through a fine needle There is only slight increase in the fixed acids, generally in the neighborhood of three millimoles per liter which is within the normal diurnal variation.

Now I suppose what you are all interested in is not how many animals we have successfully operated upon but how many humans we have operated upon. The details of our animal experiments have appeared in two articles recently published The human patients we have operated upon, using the apparatus, have not yet been reported.

We have used the apparatus to carry temporarily the entire cardiorespiratory functions of four human patients The first was operated upon a year and one-half ago, and the last in July, 1953, of this year. We have one surviving patient who is quite well in every way with complete closure of an atrial septal defect The three deaths have all been due to human error and not to failure of the apparatus

The first patient, who was operated upon a year and one-half ago, was a fifteen-month-old baby that weighed eleven pounds and was in severe congestive cardiac failure Attempts at cardiac catheterization in this baby were unsuccessful It was the opinion of everyone who saw this baby that the cardiac abnormality was an interatrial septal defect We explored the right side of the heart using the apparatus and discovered that no atrial septal defect existed The child died after operation and at postmortem was shown to have a huge patent ductus arteriosus which had not been recognized at the time of operation This, of course, illustrates the importance of complete exploration of every heart which is operated upon We might have saved this child's life if we had closed the ductus

The second patient was operated upon May 6, 1953 She was an eighteen-year-old girl who had a large interatrial septal defect proved by cardiac catheterization The patient had been symptom free until about six months before operation, when she began to show symptoms of right- sided heart failure She was hospitalized three times in these six months Every time she returned to ordinary activity, she had symptoms of heart failure Cardiac catheterization revealed an atrial septal defect with a left to right shunt through the defect amounting to nine liters per minute The patient was connected with the apparatus for forty-five minutes and for twenty-six minutes all cardiorespiratory functions were maintained by the apparatus She had a large interatrial septal defect which was quite easily closed with a continuous silk suture The patient's postoperative convalescence was uneventful She was readmitted to the hospital in July, 1953 At this time, cardiac catheterization showed that the septal defect was completely closed and that there was no evidence of any shunt The cardiac murmur had completely disappeared and she was in good health We believe that a transverse incision extending from one axilla to the other opening both pleural cavities through the fourth interspace, and dividing the sternum, gives the best exposure for this type of cardiac operation The chest wound heals quite solidly and results in an inconspicuous scar beneath the breasts

The last two patients were operated upon in July, 1953 They were both underdeveloped girls aged five and one-half years Each of them weighed only about thirty pounds The first child had a large interatrial septal defect proved by cardiac catheterization Cardiac arrest occurred after we had opened the chest but before we had cannulated any vessels The heart became blue and dilated as the chest was being explored. We tried for one hour to establish normal cardiac contractions but were unable to do so We then rather reluctantly cannulated the superior and inferior venae cavae and the left subclavian artery while an assistant massaged the heart As soon as the patient was connected with the apparatus the heart action became strong and the color of the heart pink We then opened the right atrium and repaired five separate defects in the interatrial septum After the defects were closed and the heart wound sutured, the ligatures around the venae cavae were cut allowing the heart to take over part of the circulation Whenever we stopped the artificial support by the machine, the heart dilated and began to fail Partial support by the extracorporeal blood circuit was maintained for three or four hours at the end of which time the cannulas were withdrawn and the chest was closed. The patient's heart, however, dilated and cardiac arrest occurred Death, of course, in the patient cannot be attributed in any way to the use of the heart-lung apparatus as cardiac failure occurred prior to the use of the apparatus Perhaps the dilatation of the heart and cardiac arrest was the result of a reversal of the shunt through the interatrial defect due to blood transfusion which was given during the early part of the operation

The second five and one-half-year-old child had a proven interatrial septal defect by cardiac catheterization It proved impossible, however, to pass the catheter into the right ventricle On clinical grounds an interventricular septal defect was thought to exist in addition to the interatrial defect It was known that the patient had a left superior vena cava which was somewhat larger than the right superior vena cava The child turned out to have not only a huge interatrial septal defect but also a large interventricular septal defect and a small patent ductus arteriosus Cannulation proceeded normally in this child and when we opened the right atrium we found it to be flooded with bright red blood returning to the atrium through the tricuspid valve Ae we could not get a clear field to work and the flow of bright red blood was so excessive, we closed the atrium and removed cannulae The child died after the operation, which was to be expected due to our failure to correct any of the cardiac defects

(Motion picture shown)

This motion picture was taken during the course of the operation in which the five separate interatrial defects were sutured. It illustrates the value of being able to visualize the interatrial septum. It seems to me that the four smaller defects might have been missed by the employment of indirect blind methods The film clearly shows how simple it is to keep the operative field clear of blood and that the heart appears of normal color and is beating well because the myocardium is receiving oxygenated blood from the apparatus through its coronary vessels

In conclusion, I would like to say that I think the work I have reported is some of the early work in this field and that there is considerably more work to be done It seems to me that there will always be a place for an extracorporeal blood circuit because it permits a longer safe interval for opening the heart than can ever be obtained by any of the hypothermia methods.

Discussion

Dr F D Dodrill, Detroit, Michigan—You have just heard an excellent review of the mechanical heart work by the leading pioneer in this country Dr Gibbon has had vastly more experience than the rest of us in this field, and I am envious of his vast accomplishments in this work

When I first began this work several years ago, it was not definitely known, nor is it now definitely known, just how this can best be done It was originally thought by those working in the field one could not bypass either side of the heart without completely bypassing the entire heart and lungs at the same time As time has gone on and various workers have made definite contributions along this line, it is now apparent that either the right side or the left side of the heart may be bypassed while the opposite side of the heart and lungs continue to perform their functions

We began a few years ago, therefore, to explore the possibilities of the following types of procedures (1) bypass of the right heart, (2) bypass of the left heart, (3) bypass of both sides of the heart using the lungs for oxygenation, and (4) bypass of the heart and lungs using the mechanical oxygenator We have performed more than 100 operations on the experimental animal doing one of these various procedures.

Our apparatus is so constructed that it can perform any or all of these functions Insofar as the mechanical heart itself is concerned, I feel there are three factors which are important These are (1) it should be strong, sturdy and not subject to breakdown, (2) it should be so constructed that all parts coming in contact with the blood can be sterilized by ordinary autoclaving, and (3) it should produce and maintain a pulsatile flow in the vascular system

Bypass of the Right Heart—Our experimental work in bypass of the right heart has been highly successful in the experimental animal This consists in taking the blood from the superior and inferior vena cava, passing it through the right side of our mechanical heart and back into the pulmonary circuit through the artery to the right lower lobe of the lung By this method, the right side of the heart can be bypassed while the opposite side and the lungs continue to perform their functions Also, if one wishes, the blood may be taken from the right atrial appendage By this method not only the superior and inferior vena cava flow, but the coronary sinus flow, as well, can be completely diverted from the right side of the heart These connections depend upon the type of procedure and the exposure which one is attempting

In our experimental work on the right-sided bypass, we encounter severe hypotensive reflexes These hypotensive reflexes are greatly increased by even the slightest degrees of anoxia This reflex has been referred to as the Bainbridge reflex but it may be possible, that additional reflexes may be present It is my impression that a great deal of physiological investigation is needed along this line On the left side, however, these hypotensive reflexes are much less acute and oftentimes in the experimental animal we see no evidence at all of a decrease in the blood pressure during a left-sided bypass. Using such a procedure, any purely right-sided heart disease may be exposed and corrected under direct vision The examples of such conditions are pure pulmonary stenosis with an intact cardiac septa and diseases of the tricuspid valve also with an intact septa Figure 1 illustrates the pulmonary valve in a patient with pure pulmonary stenosis during a right-sided bypass This valve was exposed for a period of twenty-five minutes while the mechanical heart pumped approximately 4½ liters of blood per minute The left side of the heart continued to perform its function as did the lungs While the right-sided bypass of the heart seems to be a rather practical procedure, it should be pointed out that there are not many pathological conditions of the right side of the heart alone in which such a procedure can be used The great majority of patients with pulmonary stenosis have a defect in either the intratrial or interventricular septum and in such a patient a unilateral bypass is, of course, out of the question There are rare instances of large emboli lodging in the right side of the heart which give some warning before death and it is possible that such large emboli could be removed during a right-sided bypass

A right-sided bypass is a practical procedure for patients with pure right-sided disease There is no need at all for bypassing the lungs as well for exposure of a purely right-sided defect The main advantage of using the mechanical lung along with the mechanical heart is that the connections of the apparatus to the anatomical structures are much easier However, with a right-sided bypass, the connections are likewise easy and should present no obstacle at all to the operating surgeon Of all these various procedures which can be done in the way of

Fig 1 Stenotic pulmonary valve in a seventeen-year-old boy

Fig 2 Open left strium exposing the mitral valve in a fifty-year-old woman

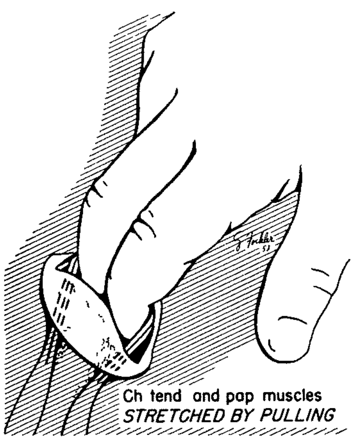

Fig 3 A method of elongating the chordae tendineae in a patient with mitral regurgitation

bypassing the entire heart or a portion thereof, I strongly feel that a right-sided bypass alone is a practical procedure which will be useful in years to come The pathological conditions, however, are rather limited for this type of a procedure

Bypass of the Left Heart—A great deal of animal experiments have been done on the left-sided bypass This consists in taking the blood from the pulmonary veins, passing it through the left side of the mechanical heart and back into the aorta which is its ultimate destination The experimental work on dogs is extremely gratifying We have been able to expose the mitral valve in dogs for fifteen consecutive times without a fatality During the left-sided bypass, the right side of the heart and the lungs continue to perform their functions During such a procedure, it is necessary to maintain a sufficient systemic blood pressure and preferably to maintain it by a pulsatile flow The hypotensive reflex which is rather acute on the right side, especially if there exists any degree of anoxia, is not so acute on the left side In numerous animal experiments, we have not encountered such a reflex at all We have applied this procedure to a few patients Figure 2 illustrates the open left atrium with the mitral valve in the depths of the cavity This patient made an uneventful recovery and is much improved from the surgical procedure It seems logical to assume that if the mitral valve can be exposed, there is more apt to be an opportunity to improve valve function especially in mitral regurgitation Figure 3 illustrates the elongation of the chordae tendineae in a patient with mitral regurgtation during a left-sided bypass Although, the valve and the chordae were not visualized, the arterial pressure pattern shows conclusively that the mechanical heart was completely maintaining the systemic circulation It has now been over one year since this patient was operated and he is vastly improved, even to the point of almost a complete disappearance of his loud murmur One additional patient has also been treated in a similar manner with encouraging results There are several factors in the production of mitral insufficiency, however, the most important one seems to be the shortening of the chordae tendineae and sometimes the papillary muscles as well The valve leaflet itself may be nearly normal, and if the chordae can be lengthened to the point where it permits valve closure, the regurgitation is markedly diminished or corrected Oher types of procedures have been done·during a left-sided bypass The aortic valve may be manipulated by inserting a finger into the left atrial appendage down through the mitral valve and upwards through the aortic valve A patient with aortic stenosis has been so treated and the valve was easily fractured with the finger Whether or not this procedure will be better than the method of going directly through the left ventricle remains to be seen

Bypass of Both Sides of the Heart—A bypass of both sides of the heart using the lungs for their natural functions is a more difficult procedure from the technical standpoint We have been able to do this, however, quite satisfactorily in experimental animals and have applied it to one patient Both sides of the heart were completely taken over by the mechanical device A stenotic pulmonary valve was corrected and the interventricular septum was exposed Unfortunately, this patient had, in addition to the pulmonary stenosis, marked infundibular stenosis and could not be sufficiently corrected During the course of the procedure, the systemic blood pressure as well as respiratory function were quite satisfactory Unfortunately, the patient succumbed on the fourth postoperative day from atelectasis and pneumonitis

A Combination of Hypothermia and the Mechanical Heart—These various procedures which I have shown you have been done solely with the mechanical heart We are now working along the lines of the combination of hypothermia and the mechanical heart Whether or not various types of intracardiac surgery will be done under hypothermia alone, using the mechanical heart alone or a combination of these two methods remains to be seen The advantage of combining hypothermia with the mechanical heart is that during hypothermia, the circulating blood volume is markedly reduced and the work of the mechanical heart is vastly decreased Moreover, with the circulation of such a small volume of blood, only one lobe of the lung needs to be used for respiration Such a small quantity of blood can be easily circulated through one lobe of the lung while the rest of the lungs are at rest This vastly minimized the technical difficulties with the use of the mechanical heart alone It is easy to cannulate one of the upper pulmonary veins, to pass the blood through the left side of the mechanical heart and back into the aorta The right-sided connections for such a procedure are similar to a complete bypass of the heart and lungs and consists simple of cannulating the superior and inferior vena cava and passing the blood back into the pulmonary circuit The lung is a very intricate organ It is not known but that it may perform other functions aside from the uptake of oxygen and the release of carbon dioxide It seems as though a good hard attempt should be made at using the lung for its natural function before we completely discard it

---

Dr Clarence Dennis, Brooklyn, New York—After Dr Maurice D Visscher and Dr Owen H Wangensteen jointly suggested to me, late in 1945, the undertaking of a project to develop an artificial heart-lung apparatus, I very quickly became aware of the activities of Dr John H Gibbon, Jr, and his group in Philadelphia I consulted him at that time, and was very cordially received, shown protocols, shown the apparatus as it had developed up to that time, and welcomed into the group of those working on this project Between that time and this, there have been associated in the research project with Dr Karl E Karlson and me approximately eighteen physicians Work has been carried out first at the University of Minnesota and more recently at the State University of New York, and has been supported by the United States Public Health Service, the Graduate School of the University of Minnesota, the State University of New York, the Life Insurance Medical Research Fund, and private sources In our search for an answer to this problem, my associates and I have come to have the very highest regard for the ingenuity, integrity, and cordiality of Dr Gibbon and his associates Success could not have come to finer people

Dr Gibbon has very carefully listed the qualifications of a satisfactory artificial heart-lung apparatus We have

Fig 1 Closure of inter-atrial septal defect, May, 1951 Closure was easily performed with interrupted silk sutures under direct vision The patient suffered air embolism before opening the atrium, a consequence of human error in failure to switch on an automatic control mechanism She died of complication of air embolism after several hours of partial support by the machine

been concerned with an additional qualification, namely, the ability to render the whole apparatus absolutely free of bacterial contamination The reason for our preoccupation with this qualification arises from the work of Dr Russell M Nelson, who was a member of our research group three years ago Dr Nelson showed that our apparatus at that time was contaminated with the paracolon bacillus, and that injection intravenously into normal dogs of a re-suspended fourteen-hour culture of this organism resulted in a symptom complex which had been responsible for most of the deaths prior to that time This symptom complex developed rapidly and killed some dogs in less than three hours The affected dogs showed a rapidly progressive metabolic acidosis, an incoagulable state of the blood, development of effusions into the serous cavities, massive gastrointestinal hemorrhage, and death in shock [3] Satisfactory sterilization of the apparatus removed this symptom complex from our list of problems The machine which we formerly used could not tolerate autoclaving, and sterilization by formaldehyde proved fairly regularly to be incomplete Nevertheless, it was possible for us to do a long series of animal perfusions with this old apparatus, with no more than a 10 per cent mortality attributable to the perfusion *per se* [1]

We gained so much confidence with this apparatus that, in conjunction with Dr Richard Varco, we perfused two patients in efforts to achieve satisfactory closure of inter-atrial defects [2] We were unsuccessful in salvaging either of these patients, but it was the consensus that the failure was not due to any intrinsic defect in the apparatus, but rather to human errors The repair which was accomplished in the second patient is shown in Figure 1 This patient was lost as a consequence of human error, namely, failure to turn on the automatic control apparatus, which had been designed and proven adequate to maintain the desired blood level in the reservoir of the oxygenator Air embolism oc-

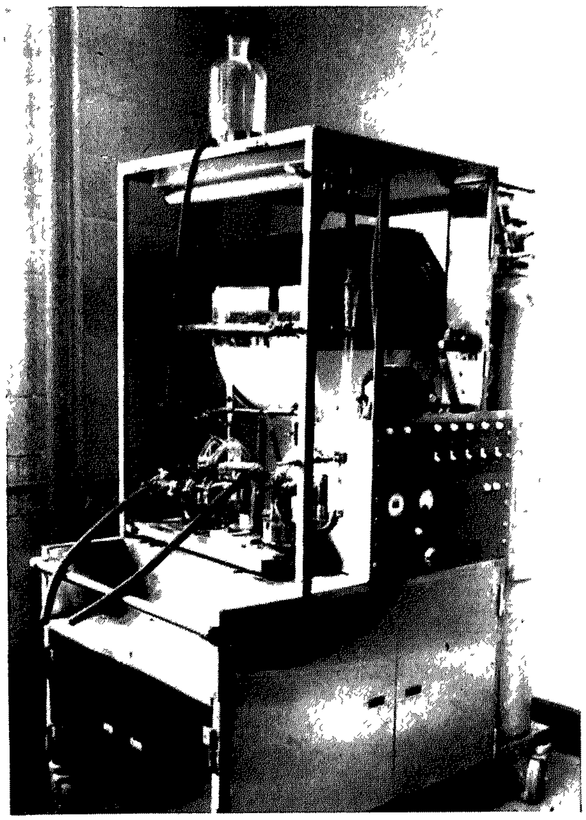

Fig 2 The present pump-oxygenator apparatus The entire blood-bearing unit can be autoclaved after assembly Oxygenation is accomplished by filming on steel mesh discs which are mounted on a rotating horizontal shaft The pumps are a modification of the Dale-Shuster pattern Risk of human error has been reduced by use of largely automatic controls

curred because of this oversight The patient survived approximately eight hours after the procedure, but only by virtue of partial support by the pump-oxygenator, and the closure at the time of autopsy was perfectly adequate Dr Gibbon's observation is in agreement with ours, namely that the margins of most inter-atrial septal defects in clinical patients are thick, and firm, and silk closure, therefore, is not usually difficult of accomplishment Closure in this instance, in the summer of 1951, was very easily accomplished

Because of our concern over the occurrence of human errors and because of our concern with regard to absolute sterility, we have spent the past year and a half in the construction of a new apparatus, the blood-bearing portion of which can be assembled and autoclaved as a unit This apparatus is not yet complete, but has reached a stage of construction which has permitted us to employ it for approximately two dozen trial perfusions The apparatus in question is indicated in Figure 2 In this apparatus, oxygenation is accomplished by the filming of blood near the center of a 50 cm disc of stainless steel mesh which is mounted on a slowly rotating horizontal shaft It will oxygenate 500 to 600 cc of blood per minute from half saturation to full saturation and has a volume content of approximately 65 cc of blood at any given moment The pumps are a modification of the Dale-Shuster pattern They are activated by hydraulic pressure, which is produced by a mechanical cam and bellows arrangement below the surface of the table The apparatus as a whole will handle about 5 liters of blood per minute It has been calculated that it should be capable of adding in excess of 350 cc of oxygen per minute and in perfusions in large dogs has been measured to add 210 cc per minute We now regularly observe sterile blood cultures at the end of perfusion, and hemolysis produces approximately 1 mg per cent of plasma hemoglobin per minute of perfusion

There are many problems which remain and which are the source of considerable concern at the present time It has been found that our greatest problem is that of postoperative hemorrhage A technique of protamine titration, which has been employed over a period of several years in our laboratory, indicates that the addition of protamine to the animal is effective for only an hour or two and that additional protamine thereafter must be added until approximately six hours have passed The platelet count is not appreciably lowered, and this, therefore, is not the reason for our hemorrhagic difficulties It is suspected that there is excessive trauma to the blood which may be responsible for the degree of hemorrhage which we have experienced, but the low plasma hemoglobin concentrations cast a doubt even on this suggestion Preliminary studies suggest fibrinolysin as the offender.

Perhaps our major remaining problems with the new machine rotate around the completion of satisfactory automatic controls We have, at times, utilized the suggestion of Professor Dogliotti of Turin, Italy, that the pumping of blood from the machine to the arterial system of the subject be governed entirely by the arterial blood pressure of the subject At other times, we have maintained a constant volume in the extra-corporeal circuit The latter appears to be the preferable arrangement As far as controls are concerned, the removal of blood from the venae cavae has been simplified by setting the apparatus up in such fashion that the pumps fill by gravity from the vena cava

There remain several metabolic problems that have to be studied An occasional perfusion is characterized by a profound loss of circulating plasma sodium An occasional perfusion is also complicated by development of marked metabolic acidosis, even when there is a negative blood culture Finally, there is occasionally a profound disappearance of circulating protein, which, as yet, we have not been able satisfactorily to explain

It is likely that Gibbon has better solutions to many of these problems than we, but all of them must be painstakingly resolved before utilization of pump-oxygenators in clinical surgery can become general

References

1 Development of a pump-oxygenator to replace the heart and lungs; an apparatus applicable to human patients, and application to one case Dennis, Clarence, Spreng, Dwight S, Jr; Nelson, George E, Karlson, Karl E, Nelson, Russell M; Thomas, John V; Eder, Walter P, and Varco, Richard L Ann Surgery, 134:709 (Oct.) 1951.

2 Acute metabolic changes associated with employment of a pump-oxygenator to supplant the heart and lungs Dennis, Clarence, Spreng, Dwight S Jr; Young, LaVonne, Nelson, George E; Karlson, Karl E., and Pereyma, Constantine Pages 165-171 in Surgical Forum Clinical Congress of American College of Surgeons Philadelphia · W B Saunders Company, 1952

3 Nelson, Russell M Metabolic effects of paracolon bacteriemia Ann Surgery, 134 885 (Nov.) 1951.